

Redirected Walking vs. Omni-Directional Treadmills: An Evaluation of Presence

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Figure 1: Left: Locomotion via Redirected Walking, Middle: Locomotion via Virtuix Omni, Right: Virtual environment with green path.

ABSTRACT

Omni-Directional Treadmills (ODT) and Redirected Walking (RDW) seem suitable for eliciting presence through a full-body walking experience, however both present unique mechanisms that can affect users' presence, comfort, and overall preference. To measure this effect, we conducted a counterbalanced within-subjects user study with 20 participants. Participants wore a wireless VR headset and experienced a tour of a virtual art museum, once using RDW and another time using a passive, slip-based ODT. Both solutions elicit similar amounts of presence, however RDW is perceived as more natural and is the preferred choice of the participants.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual Reality

1 INTRODUCTION

Many applications of virtual reality involve navigating large virtual environments (VE). While allowing users to walk naturally enhances their sense of presence the limits of the physical walking space would restrict users from exploring larger environments [6]. This has led to various proposed solutions aiming to overcome the physical space limitations, while also preserving different benefits of natural walking. Of the proposed solutions, we are particularly interested in passive, slip-based Omni-Directional Treadmills (ODTs) and Redirected Walking (RDW) as we believe these locomotion methods to have the most body engagement and want to understand how these specific modes of body engagement may affect presence differently. Thus, we are interested in investigating the following research questions:

RQ1 Of the two locomotion methods, which results in a higher sense of presence?

RQ2 Of the two locomotion methods, which is preferred by participants?

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RQ3 How do the limitations of steer-to-center RDW and passive, slip-based ODT influence presence?

From our user study, we expect to find RDW to provide a higher sense of presence than ODT and for participants to generally prefer RDW over ODT. In addition, we expect participants will cite a fear of leaving the walking space and the reset experience as contributing to their dislike for RDW while discomfort to contribute to participant's dislike of ODT.

2 MATERIALS AND METHODS

We conducted a counterbalanced within-subjects study with two conditions where participants completed a guided art museum tour in VR. To move, participants used the Steer-To-Center algorithm for RDW with Face-To-Center resets as implemented by the Redirected Walking Toolkit [2] in a 6m x 6m space and the Virtuix Omni¹. The Omni is a passive, 'slip-style' ODT which differs from active ODTs closely resembling traditional gym treadmills [4]. The study protocol was approved by the University's Institutional Review Board.

2.1 Participants

20 participants were recruited through open calls sent through university email lists, class announcements, and other university information distribution venues. 18 were male and 2 were female, ages 18-25. 4 participants had never used a VR headset before, 12 participants played video games regularly (at least twice a week), and 19 participants were graduate or undergraduate students.

2.2 Virtual Experience

We used the MuseumVR asset available from the Unreal Engine Marketplace² and included ambient museum noise³ to the VE to simulate the ambience of people walking in a museum. The audio track included footsteps and light, indistinct chatter. In addition, we added a pre-recorded voice-over to give instructions during the experiment.

To navigate through the environment, all participants were asked to follow a green path drawn on the floor of the VE as seen in Fig. 1. The path was identical across all participants and study conditions. The path guided participants through a sequence of points of interest in the form of art pieces, spanning four rooms.

¹<https://virtuix.com/>

²www.unrealengine.com/marketplace/en-US/product/museum-vr

³www.youtube.com/watch?v=bWDbTyEEWjs



Figure 2: Reported measures for the two experimental conditions: (a) igroup Presence Questionnaire (IPQ) (*higher is better*), (b) Cybersickness Questionnaire (CSQ) (*lower is better*), and performance metrics. Error bars indicate the standard error, and the labeled whiskers indicate significant pairwise comparisons (* $p < 0.05$)

2.3 Procedures

After providing written informed consent, each participant was guided to the play space and donned the required technology for the given condition. After each tour completion, participants sat down at a desktop and completed a post-experience questionnaire which included the igroup presence questionnaire (IPQ) [1] and cybersickness questionnaire (CSQ) [5]. Then, participants were given a 5-minute break and encouraged to hydrate. After, participants repeated the tour and questionnaire for the remaining condition. The order of experimental conditions was counterbalanced. At the end, participants completed a post-study questionnaire which included demographics and prompted participants to choose either locomotion method as the most comfortable, least distracting, and recommendation for locomotion solutions. While completing the tour, the program logged task completion time which was later processed to report the average time to target (ATT), total time in tutorial (TT), and in game (TG) for each locomotion method and participant.

3 RESULTS

The results for IPQ, CSQ, and performance metrics are shown in Fig. 2. We analyzed the responses and performance metrics with one-way ANOVAs or ANCOVAs to determine a statistically significant difference between the two locomotion methods on the dependent variables while controlling for the demographics age, VR usage, or video game usage. Regarding the two factors reported from the CSQ, a one-way ANOVA showed that ODT resulted in significantly higher dizziness scores compared to RDW, $F(1,39)=21.72$, $p<0.001$, $\eta_p^2=0.591$. However, a one-way ANCOVA showed RDW resulted in significantly higher difficulty focusing scores compared to RDW after controlling for the effect of video game usage, $F(1,39)=7.18$, $p<0.017$, $\eta_p^2=0.323$. Regarding the four factors reported from the IPQ, a one-way ANCOVA showed that spatial presence scores were significantly higher for RDW than for ODT after controlling for the effect of VR usage, $F(1,39)=8.65$, $p=0.01$, $\eta_p^2=0.366$.

For RDW, the average total reset count was 27.85 with a reset triggered on average every 23.91 seconds. The average time needed to perform a reset was 5.73 seconds and participants spent 70.68% of their time in the tour outside of the reset state.

In analyzing participants' response to which locomotion methods they found to be the most comfortable and distracting and which they would recommend as a locomotion solution we found that participants felt RDW to be more comfortable despite the repeated turning due to the resets. RDW felt more natural and allowed for more freedom of movement. However, for RDW participants reported a worry of tripping, falling, or colliding with the real world. Conversely, participants cited the safety of the ODT as the main benefit of the system. The movement in the ODT felt less like walking and more

like running or skating. This along with feelings of being constricted took away from their reported sense of presence.

4 CONCLUSION AND FUTURE WORK

As anticipated, the semi-natural motions afforded by an ODT can be undesirable for many users. This led to frustration due to a mismatch between users' intended movements and resulting motions. With RDW, users alternated between periods of natural walking and turning in place for resets. Intuitively, the walking portions are desirable while the resets are not. With the space constraints and algorithm that we employed, participants spent an average of 30% of their time in a reset state, which was both time-consuming and disruptive. Nevertheless, for some participants the looming threat of leaving the walking space also brought a degree of unease.

This work is capturing a moment in the state of accessible RDW and ODT options for end users. With the use of more effective redirection, a reduction of resets, and improvement of the experience of resets [3], the preference for RDW can only become stronger. Conversely, the future of ODTs might involve designs that feel more comfortable and allow for easier use, which could keep these contending approaches as ongoing rivals for years to come.

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